

Internship Project Report

Physical Design Implementation Flow

Team 1

Aavanthika M Nair

Jacob Saji

Hrishkeesh Narayan P E

Contents

1	Project Overview	3
2	Design Methodology	3
2.1	Functional Simulation	3
2.2	Synthesis	4
2.3	Floorplan and Powerplan	4
2.4	Placement and Clock Tree Synthesis (CTS)	4
2.5	Routing	4
3	Results and Screenshots	4

1 Project Overview

The primary objective of this internship project was to successfully execute a complete Physical Design (PD) implementation flow for a digital integrated circuit. Physical design is a critical phase in the Application-Specific Integrated Circuit (ASIC) development cycle, bridging the gap between a logical circuit description and the final physical layout required for semiconductor fabrication.

Throughout this project, we utilized industry-standard Electronic Design Automation (EDA) tools, specifically focusing on the Cadence toolchain (including Genus for synthesis and Innovus for place-and-route). The implementation process involved transforming a high-level Register Transfer Level (RTL) design into a fully routed, optimized, and verified standard-cell-based layout.

Key goals achieved during this flow included meeting rigorous Power, Performance, and Area (PPA) targets. This required iterative optimization across multiple stages: synthesizing the behavioral code into a gate-level netlist, strategically planning the silicon floorplan and power distribution networks, placing standard cells to minimize wirelength and congestion, synthesizing a balanced clock tree, and finalizing the intricate metal routing. The project culminated in generating a layout that strictly adheres to the foundry's Design Rule Checks (DRC) and Layout Versus Schematic (LVS) constraints, ensuring the design is functionally correct and ready for sign-off.

2 Design Methodology

The design methodology strictly follows a standard ASIC physical design flow. Each stage serves as a foundation for the next, requiring careful optimization and validation before progression.

2.1 Functional Simulation

Before proceeding to synthesis and physical implementation, the RTL design was functionally verified through simulation using a Verilog testbench. The complete design hierarchy, including all individual modules, was compiled together with the testbench to validate the intended functionality.

The simulation verified that the top-level module correctly integrated the clock divider, vibration comparator, temperature comparator, decision logic, and warning generator. Different input stimulus values were applied through the testbench to emulate various operating conditions. The generated waveform confirmed that the design responded correctly to changes in sensor inputs and produced the expected warning outputs.

Performing functional simulation prior to synthesis ensured that the RTL implementation was free from logical errors, thereby reducing the likelihood of costly design iterations during the later stages of physical implementation.

2.2 Synthesis

Using Cadence Genus, the initial RTL code was translated and optimized into a gate-level netlist. This stage mapped the behavioral logic to the target standard cell library, applying early constraints to balance power consumption, operational speed, and total silicon area.

2.3 Floorplan and Powerplan

Executed in Cadence Innovus, the floorplanning stage defined the physical dimensions of the die and core area. Hard macros were strategically placed to minimize routing congestion and wire delays. Following this, a robust Power Distribution Network (PDN) was established. VDD and VSS rings, stripes, and standard cell rails were synthesized to ensure uniform power delivery across the chip and to mitigate IR drop and electromigration risks.

2.4 Placement and Clock Tree Synthesis (CTS)

The gate-level netlist components were assigned to precise physical coordinates within the core. Innovus placement engines optimized these locations to meet setup timing requirements while avoiding localized congestion. Once placed, Clock Tree Synthesis (CTS) was performed to distribute the clock signal from its root to thousands of sequential elements (flip-flops). Buffers and inverters were inserted to create a balanced, tree-like network, aggressively minimizing clock skew and controlling insertion delays to ensure robust hold timing.

2.5 Routing

The final major implementation step involved executing the physical wiring between all placed cells, macros, and I/O ports. Using the available routing metal layers and vias, signal nets were connected to meet final setup and hold timing constraints. This stage heavily focused on resolving minor timing violations and eliminating complex DRC and LVS errors to produce a clean, foundry-ready layout.

3 Results and Screenshots

The following figures illustrate the verification and implementation stages of the project. Functional simulation was first performed to validate the RTL behavior, followed by synthesis, floorplanning, power planning, placement, and routing. These screenshots demonstrate the successful progression through the complete ASIC physical design flow.

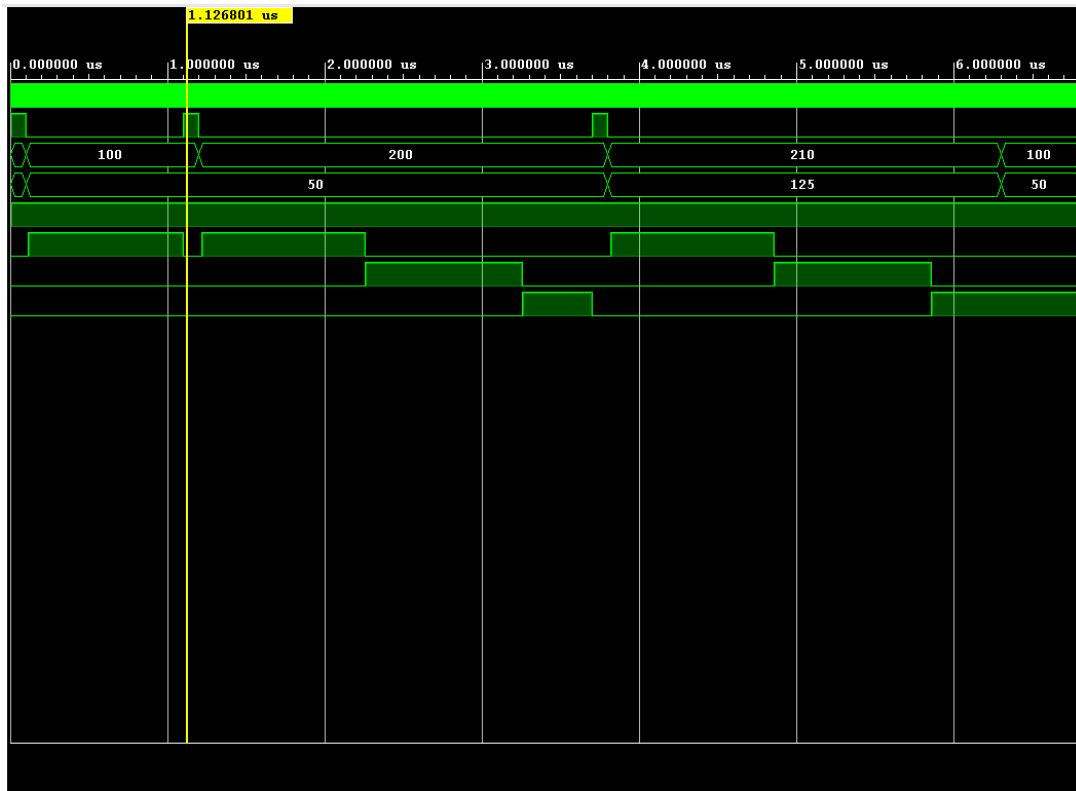


Figure 1: Functional Simulation Waveform showing correct RTL behavior.

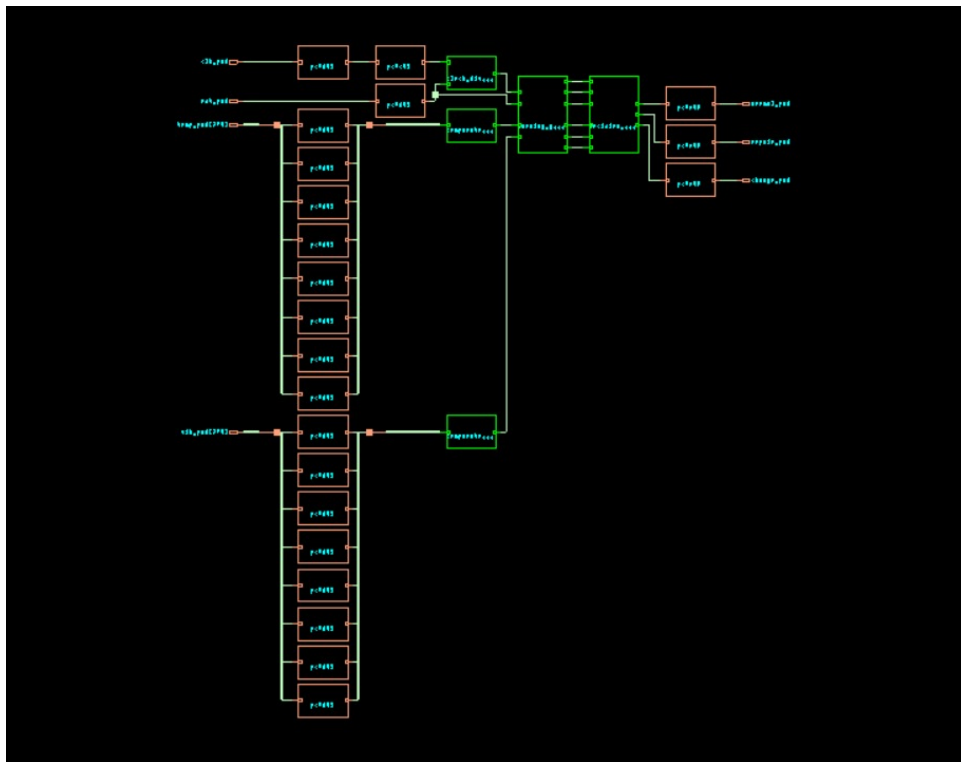


Figure 2: Synthesis Output: Gate-level netlist generation.

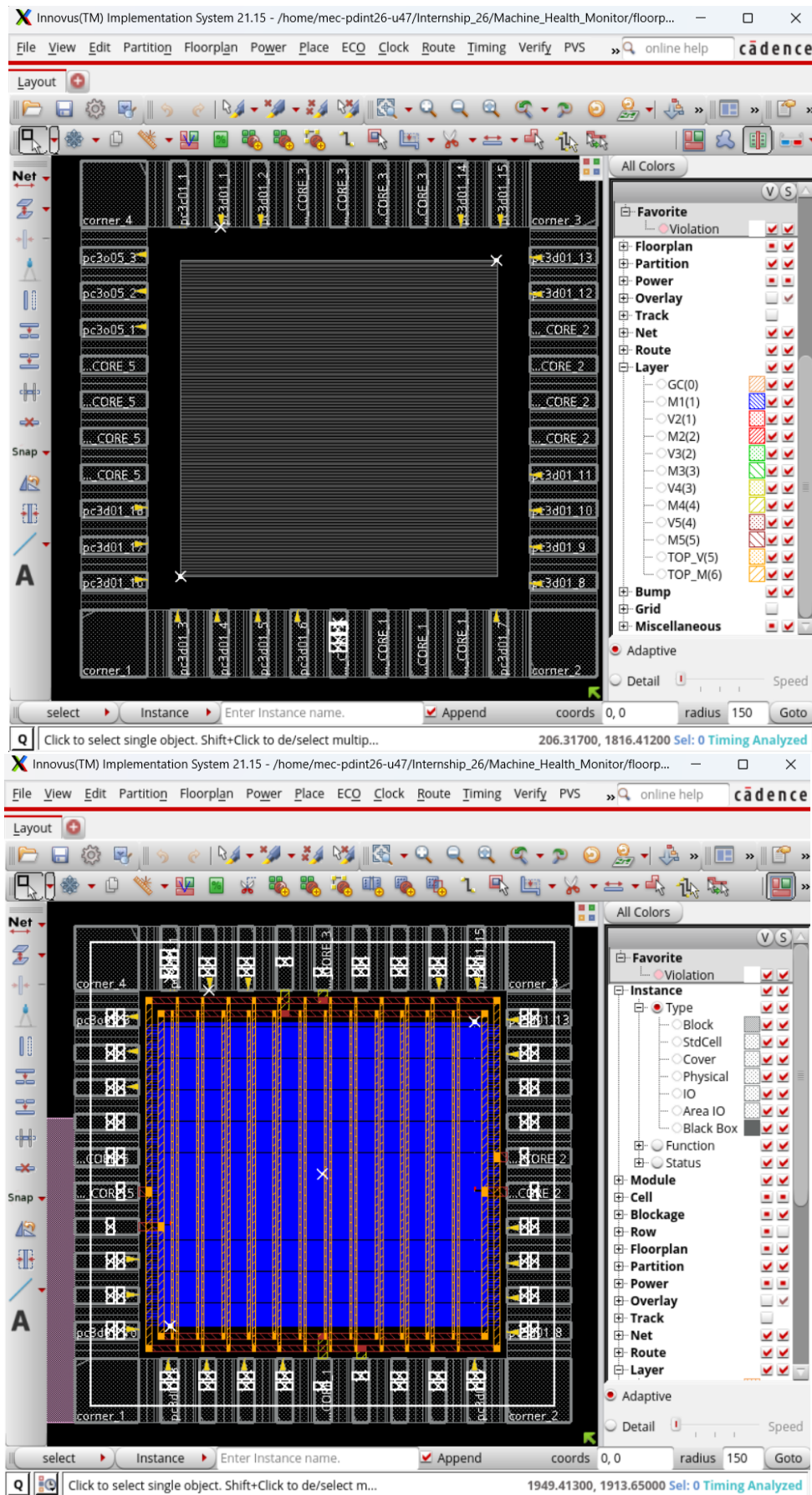


Figure 3: Floorplan and Powerplan View: Core area definition and PDN structure.

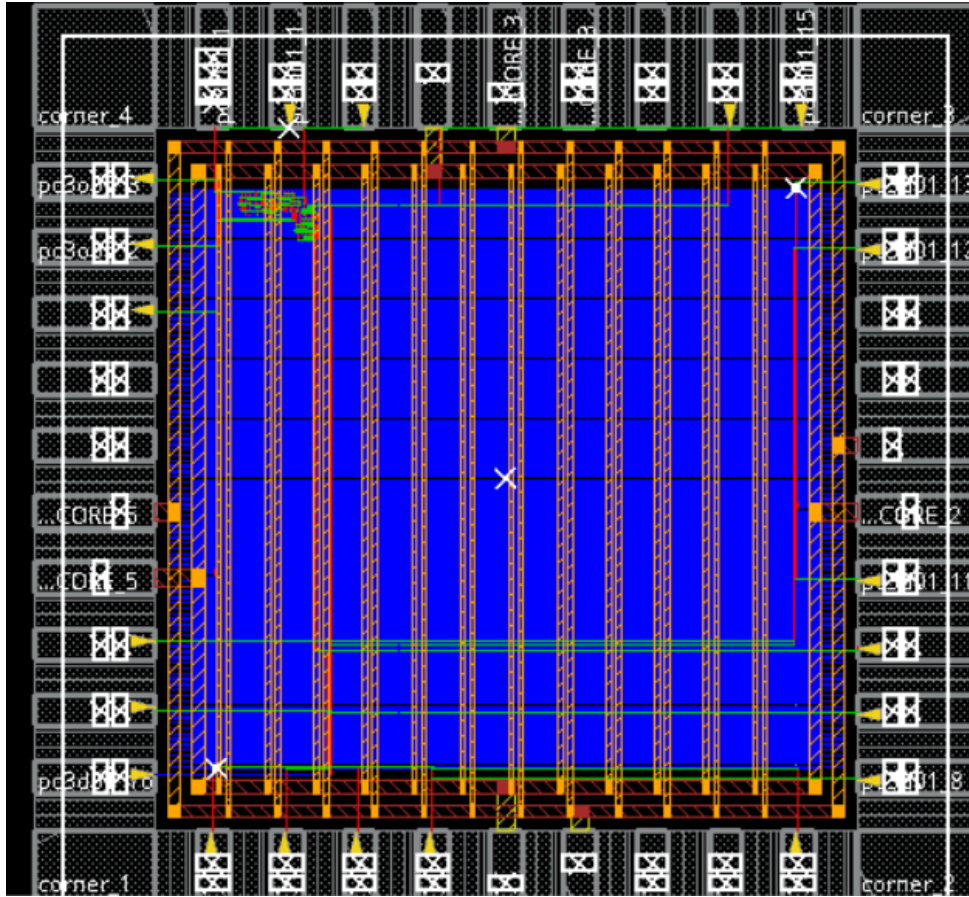


Figure 4: Placement View: Standard cell distribution within the core.

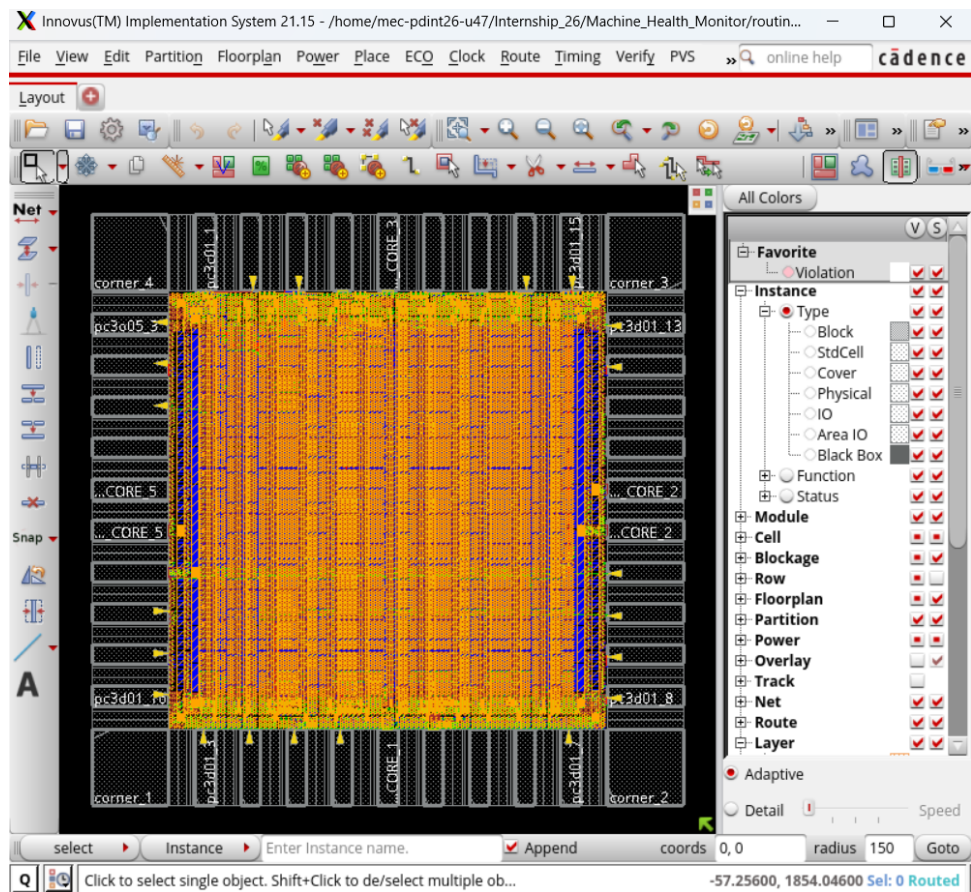


Figure 5: Final Routed Design: Completed signal routing and metal layers.